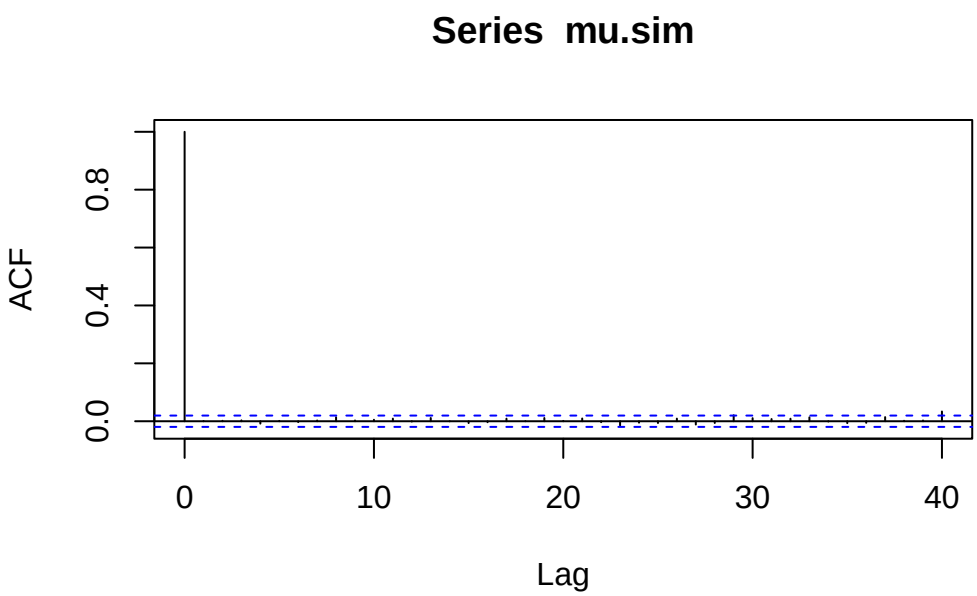
Assignment 3

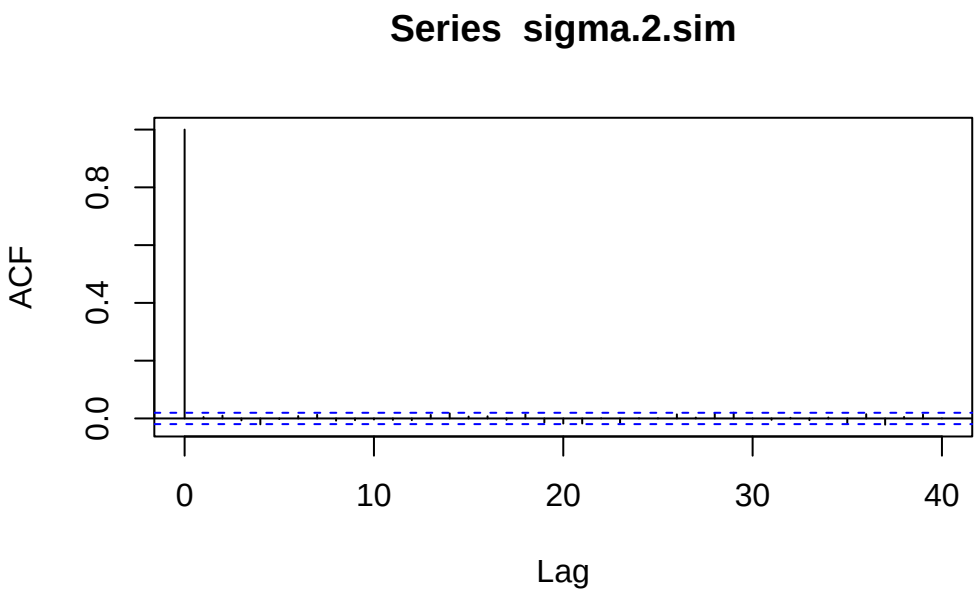
Problem 1

a. Simulates from the posteriors for μ and σ^2 and produce autocorrelation plots for their respective successive variates

```
source("./Flint Gibbs.r", echo = FALSE)
acf(mu.sim)
```




```
acf(sigma.2.sim)
```



b. Metropolis sampler

i. Find a value of `rho` that gives an average acceptance rate of about 0.35

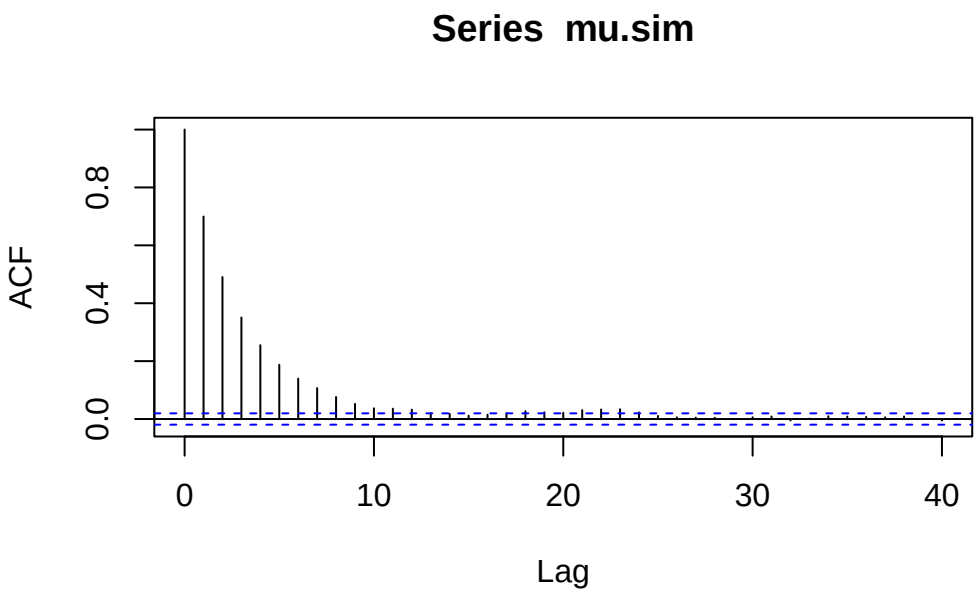
- With the `rho = 0.0308` the acceptance rate is around 0.35

```
rho <- 0.0308
source("./Flint Metropolis.r", echo = FALSE)
mean(accept.prob)
```

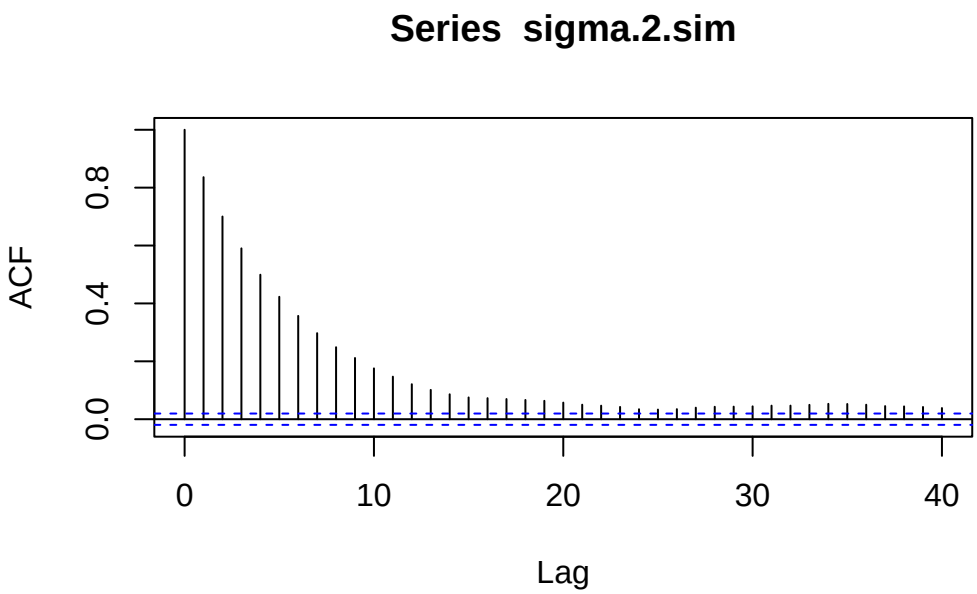
```
[1] 0.3505527
```

ii. Plot the successive autocorrelation plots for μ and σ^2

```
acf(mu.sim)
```




```r
acf(sigma.2.sim)
```



c. Which method exhibited faster mixing

- **The Gibbs Sampler** exhibited faster mixing as the autocorrelation drops within the significance bounds faster compared to the Metropolis Sampler.

Problem 2

a. Use the model in `polls20161.bug` for the following questions

i.

```
d <- read.table("./Polls 2016.txt", header=TRUE)

# variable init
d$sigma <- d$ME / 2

# JAGS
library(rjags)
```

```
Loading required package: coda


Linked to JAGS 4.3.2


Loaded modules: basemod,bugs
```

```
initial.vals <- list(
    list(mu=100, tau=0.01),
    list(mu=100, tau=100),
    list(mu=-100, tau=0.01),
    list(mu=-100, tau=100)
)

ml <- jags.model(
    "./Polls 2016.bug",
    d,
    initial.vals,
    n.chains=4,
    n.adapt=1000
)
```

```
Warning in jags.model("./Polls 2016.bug", d, initial.vals, n.chains = 4, :
Unused variable "poll" in data


Warning in jags.model("./Polls 2016.bug", d, initial.vals, n.chains = 4, :
Unused variable "ME" in data
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 7
   Unobserved stochastic nodes: 9
   Total graph size: 42

Initializing model
```

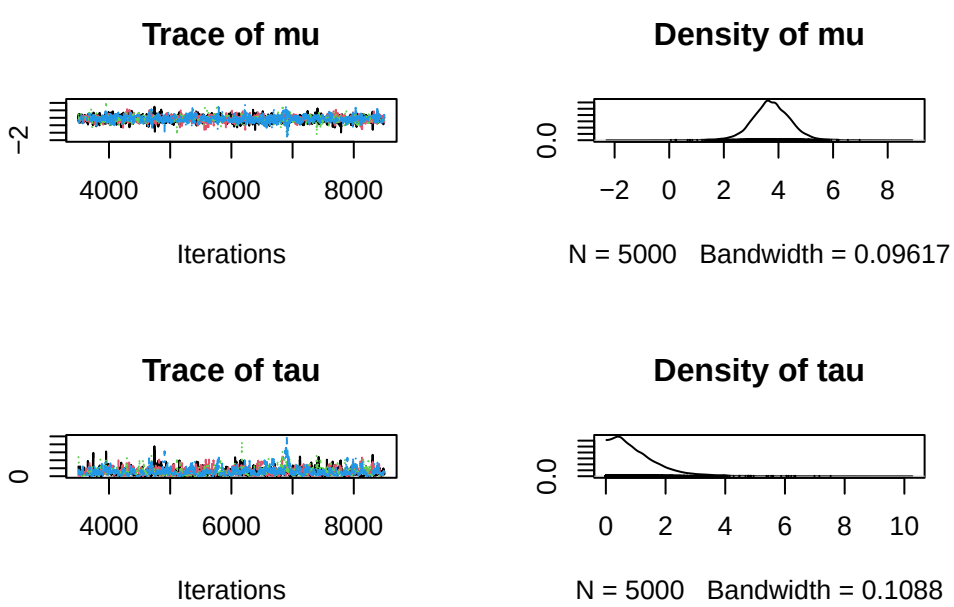
ii.

```r
update(ml, 2500) # burn-in

x <- coda.samples(ml, c("mu", "tau"), n.iter=5000)
```

iii.

- Based on the trace of `mu` and `tau`, there is no convergence problem.

```r
plot(x, smooth=FALSE)
```



iv.

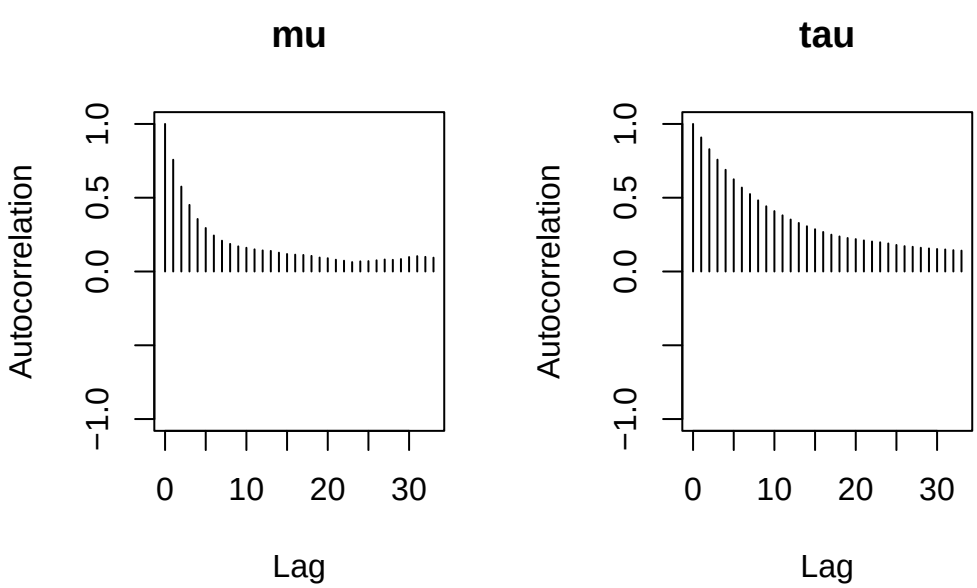
- Based on the result of the Gelman-Rubin statistics, both μ and τ have the point estimation around 1.00, which indicates a close-to-perfect convergence.

```r
gelman.diag(x, autoburnin=FALSE)
```

```
Potential scale reduction factors:

    Point est. Upper C.I.
mu        1.00       1.00
tau       1.01       1.01

Multivariate psrf

1
```

v.

```r
# extract the first chain from x
first <- x[[1]]
autocorr.plot(first)
```



vi.

- Both are considered adequate as both of them exceed the suggested minimum value of 400

```
effectiveSize(x)
```

```
       mu       tau
2204.2468  872.9859
```

b. A new model that uses an almost flat prior for `tau` on the log scale

i.

```
model {

  for (j in 1:length(y)) {
    y[j] ~ dnorm(theta[j], 1/sigma[j]^2)
    theta[j] ~ dnorm(mu, 1/tau^2)
  }

  mu ~ dunif(-1000,1000)
  logtau ~ dunif(-100,100)
  tau <- exp(logtau)

}
```

ii.

```
initial.vals <- list(
    list(mu=100, logtau=log(100)),
    list(mu=100, logtau=log(0.01)),
    list(mu=-100, logtau=log(100)),
    list(mu=-100, logtau=log(0.01))
)

ml <- jags.model(
    "./Polls 2016_2.bug",
    d,
    initial.vals,
    n.chains=4,
    n.adapt=1000
)
```

```
Warning in jags.model("./Polls 2016_2.bug", d, initial.vals, n.chains = 4, :
Unused variable "poll" in data

Warning in jags.model("./Polls 2016_2.bug", d, initial.vals, n.chains = 4, :
Unused variable "ME" in data

Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 7
   Unobserved stochastic nodes: 9
   Total graph size: 44

Initializing model
```

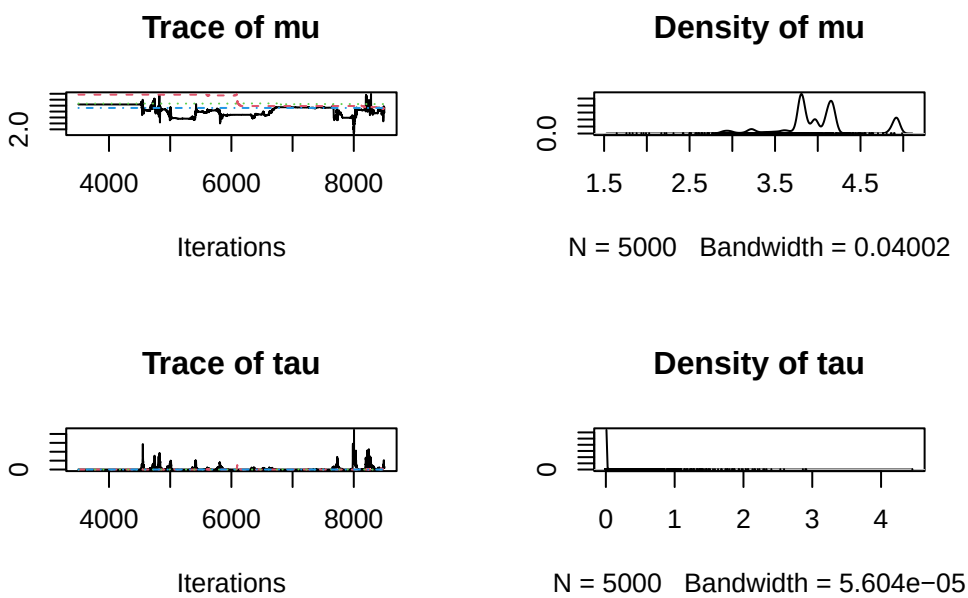
iii.

```r
update(ml, 2500) # burn-in
x <- coda.samples(ml, c("mu", "tau"), n.iter=5000)
```

iv.

- Based on the trace of `mu` and `tau`, there is a convergence problem as the mixing of both variables are poor.
  - `mu`: The chains remain in separate regions.
  - `tau`: Most values are close to 0, with occasional spikes.

```r
plot(x, smooth=FALSE)
```



v.

- Based on the result of the Gelman-Rubin statistics, both μ and τ have the point estimation greater than 1.1, which indicates poor convergence.

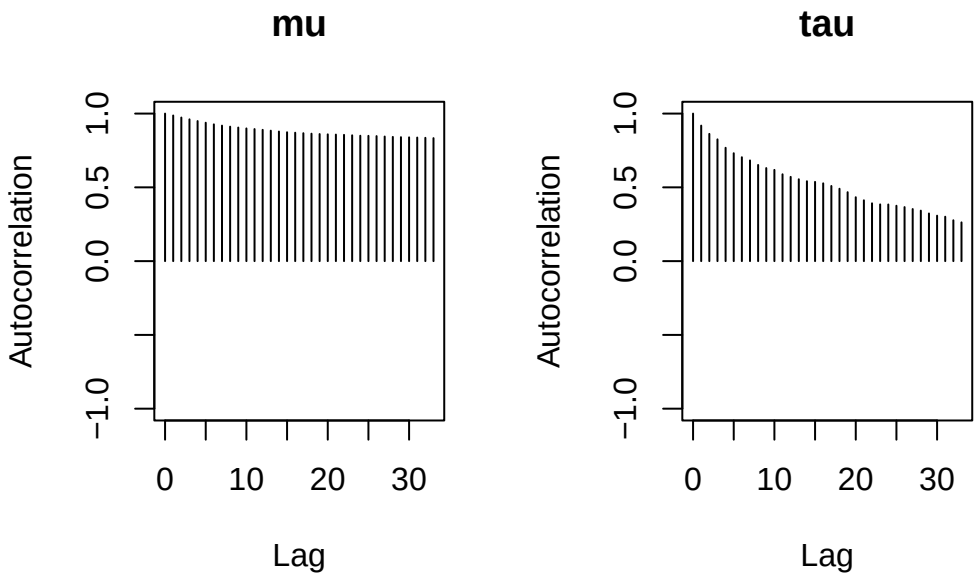
```
gelman.diag(x, autoburnin=FALSE)
```

```
Potential scale reduction factors:

    Point est. Upper C.I.
mu        1.92       4.16
tau       1.35       3.15

Multivariate psrf

1.78
```

vi.

- Based on the autocorrelation plots, the speed of mixing of both `mu` and `tau` is rather slow.

```
autocorr.plot(x[[1]])
```



vii.

- The problem with this model is that using an imporper flat prior on $log\tau$ implies $p(\tau) \propto \frac{1}{\tau}$, which is itself improper. This can lead to an improper posterior, which is not a valid probability. Such a model can cause poor convergence behavior like the statistical results above.